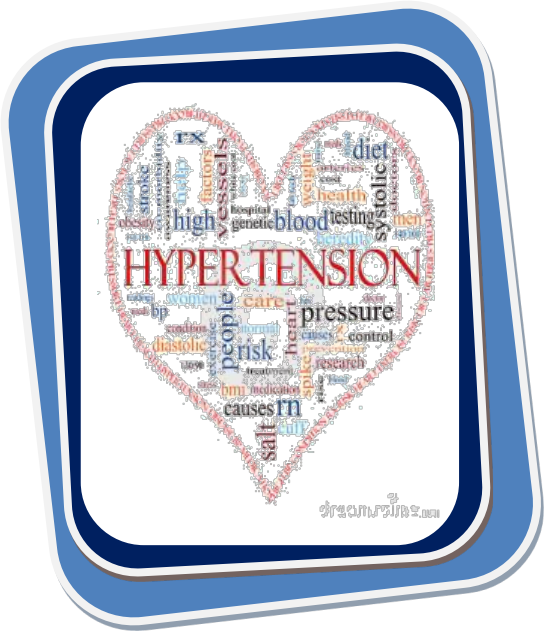
A Patient's Guide to Hypertension

Your guide to understanding high blood pressure, its causes, and treatment



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Chapter 1. Understanding Hypertension

This booklet is to help you understand high blood pressure, or hypertension, and learn the best ways to manage it.

Introduction

There are many things that you can do to help control your blood pressure and this booklet will help you understand how you can help. Keep in mind that:

1. This booklet has a lot of information. Take your time reading it a little bit each day. You will learn more each time you read it…take your time!

2. Your health care provider can answer any questions you have.

3. It will take time to make changes.

Changes Will Take Time!

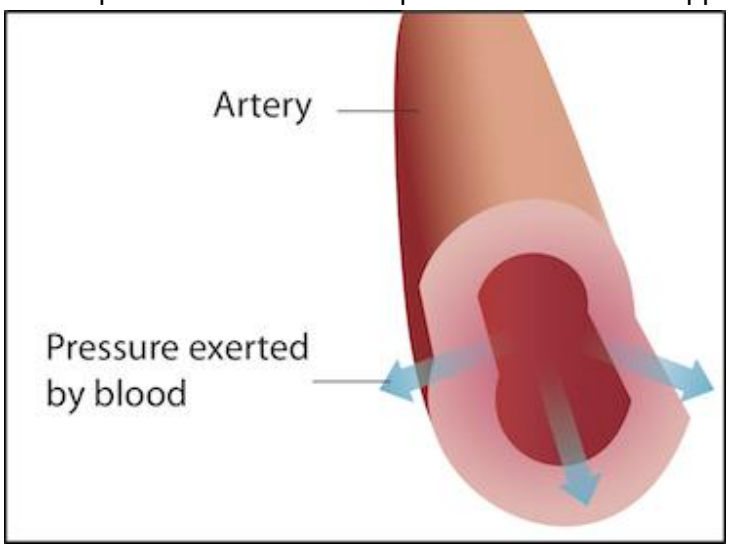
Most importantly, remember that when you work with your healthcare team to make changes with your medicines and lifestyle, you will be improving your heart and kidney health in addition to your general health and wellness.

What is hypertension?

Hypertension is the medical term for high blood pressure.

What is blood pressure?

Blood pressure refers to the pressure that blood applies to the inner walls of the arteries. Arteries carry blood from the heart to other organs and parts of the body.



Your blood pressure gives you information about how well your heart is working and what condition your arteries are in. For example, when the amount of blood your heart pumps out increases or your arteries become stiff, your blood pressure increases.

What are the symptoms of hypertension?

Hypertension does not usually cause symptoms. That is why it is important to have routine monitoring of your blood pressure to determine whether it is high.

How do I know if I have high blood pressure?

Blood pressure can be measured in your health care provider's office, at home, or even at your local pharmacy or grocery store. Blood pressure is measured with a device called a sphygmomanometer. Blood pressure is recorded in millimeters of mercury (mmHg).

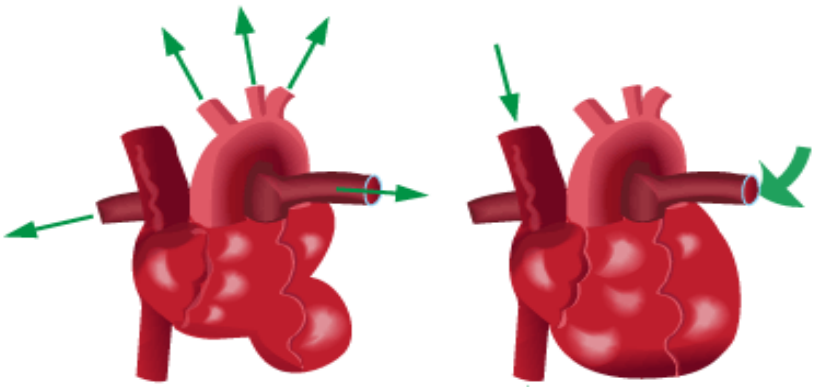
There are two types of blood pressure monitors: manual and digital.
*Manual blood pressure monitors require a stethoscope and are used by trained personnel in the office setting.
*Digital blood pressure monitors (like the picture to the left) do not require specialized training or a stethoscope. Because they are easier to operate, digital monitors are a good option for home use.

What do the numbers mean?

Your blood pressure is made up of two numbers. For example, your health care provider may tell you that your blood pressure is "150 over 90." The first (or top) number is called the systolic pressure. This is the pressure inside your arteries when your heart is contracting (at the time of a heart beat). The second (or bottom) number is called the diastolic pressure. This is the pressure inside your arteries when your heart is relaxed (between heart beats).

Blood pressure is reported as the systolic over diastolic pressure, as illustrated BELOW. Both numbers are important. As we get older, blood vessels usually become stiff or rigid, so that they are less able to dilate or expand when blood enters from the heart. Therefore, the systolic pressure (top number) usually increases with age.

<u>150</u> = this number is systolic blood pressure
90 = this number is diastolic blood pressure



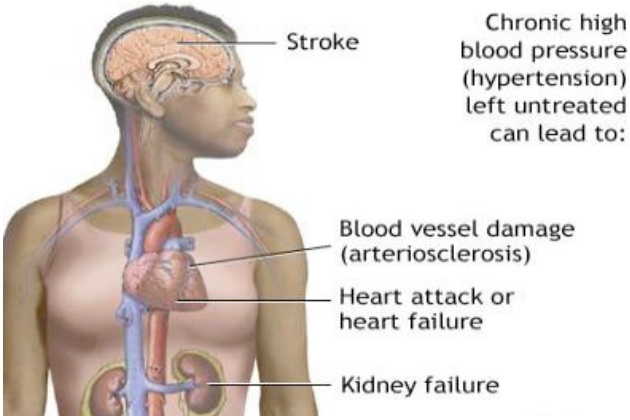
Systolic Pressure
Pressure inside your arteries when your heart is contracting (at the time of a heart beat)

Diastolic Pressure
Pressure inside your arteries when your heart is relaxed (in between heart beats)

Normal blood pressure is below 120/80. If your systolic pressure is 120 to 129, and your diastolic pressure is less than 80, you have “elevated blood pressure”. A blood pressure above 130/80 is considered hypertension, or high blood pressure. Ask your health care provider for guidelines on what a high blood pressure is for you and what your target blood pressure goal should be.

What other diseases are related to hypertension?

High blood pressure is a condition that puts you at risk for heart attack, stroke, heart failure and kidney disease. Untreated high blood pressure increases the strain on the heart and arteries, and eventually damages the heart, brain and kidneys.



What causes hypertension?

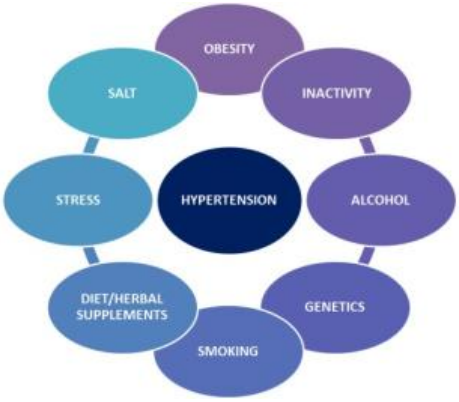
Hypertension is very common, especially as people grow older. For example, hypertension occurs in about 2 out of 3 people over the age of 60 years.

Most adults have “primary hypertension,” which means that the cause of high blood pressure is not known. A small subgroup of adults (about 5 out of every 100) has “secondary hypertension,” which means that there is an underlying and possibly correctable cause of the high blood pressure, usually a kidney or hormone disorder. Your healthcare provider will determine if you have a treatable cause for your high blood pressure.

You have (circle one):

- **Primary hypertension**
- **Secondary hypertension**

It is believed that primary hypertension is affected by genetics (heredity), obesity, high salt intake and emotional stress. The following factors sometimes also play a role in the development of high blood pressure: excessive alcohol use, smoking, sleep apnea, some herbal remedies, diet pills and other stimulants, and lack of physical activity.



Can hypertension be cured?

Hypertension usually cannot be cured. Sometimes, people who lose a large amount of excess weight, or greatly lower their use of alcohol and/or salt in their diet, and greatly reduce excess stress, may see their blood pressure return to a normal level.

How is hypertension treated?

Treatment of hypertension always includes improving lifestyle habits, including:

- **Eating the right kinds of foods**
- **Increasing physical activity**
- **Reducing salt intake**
- **Stopping smoking**
- **Losing excess weight if overweight or obese**
- **Limiting alcohol intake (no more than one drink per day for women; two drinks per day for men); a serving size is considered 12 ounces of beer, 5 ounces of wine, or 1.5 ounces of whiskey**

In addition, most patients with high blood pressure have medications prescribed by their health care provider. Following the lifestyle changes listed above can help blood pressure medicines to work better.

Important things to do every day

If your health care provider has prescribed medicine to lower your blood pressure, the most important thing that you can do is to take it as instructed. If the medicine causes side effects, do not stop taking it without talking with your healthcare provider about the side effects you are having. Your dose may be lowered, or the medicine may be changed.

If the cost of your medicine is a problem, please tell your healthcare provider. Your provider can change the medication or help you enroll in a medication assistance program to help cover the cost of the medication. Taking your blood pressure medicine can help prevent you from having a heart attack or stroke and can even save your life!

What else can you do?

In the past, only blood pressure measurements made in the health care provider's office were used to determine whether your blood pressure was under control. Now, home measurements with a battery-operated, semi-automatic device can be used to regularly measure blood pressure. Home blood pressure measurements help your health care team determine whether your medications need to be changed. Research has shown that people who measure their blood pressure at home and report the results to their health care provider have better blood pressure control.

Chapter 2. Know Your Medications

Introduction

Make a complete list of all of your medications. This includes all over-the-counter medications such as aspirin, vitamins, diet or herbal supplements as well as inhalers and insulin. Make sure to include the name of the medication, the strength (usually in milligrams) and number of tablets (dose) you take, and how often you take it. Always bring this list with you to all of your clinic appointments, emergency department visits and hospital visits. A sample medication list is in the table below. You can use the blank medication lists which is in the "Additional Information" section of this booklet.

Medication Name	Strength	Dose	Frequency
Lisinopril	20 mg	2 tablets	Every morning
Multivitamin	---	1 tablet	Every morning
Vitamin C	1000 mg	1 tablet	Every morning

Remember to take your medications!

Always remember to take all of your medications as instructed by your provider. Some people find that using a pillbox or setting a reminder on their phone helps them remember to take their medication. If you think that using a pillbox will help you remember to take your medications but you can't afford one, please let us know.

Choosing the best medication(s)

Your provider may prescribe one or more medications to help control your blood pressure. A number of factors will be considered to help determine the best medication(s) for you, including your age, laboratory studies, blood pressure goal, other medications you take, and whether you have other medical conditions such as diabetes, chronic kidney disease, or heart failure.

Common blood-pressure lowering medication(s)

Angiotensin Converting Enzyme Inhibitor (ACE-I)

Generic Name	Dose	How They Work	Side Effects
Names ends in “pril” (such as lisinopril or enalapril)	1-2 times per day	Lower blood pressure by widening blood vessels and helping blood flow easily	Dry cough and increased potassium level are the most common side effects. Kidney function and potassium levels should be checked regularly by your provider. If you have swelling of the face or throat, then stop taking the medication and seek medical help immediately.

Angiotensin Receptor Blocker (ARB)

Generic Name	Dose	How They Work	Side Effects
Names ends in “tan” (such as losartan or valsartan)	1-2 times per day	Lower blood pressure by widening blood vessels and helping blood flow easily	Increased potassium level is the most common side effect. Kidney function and potassium levels should be checked regularly by your provider. If you have swelling of the face or throat, then stop taking the medication and seek medical help immediately.

Calcium Channel Blocker

Generic Name	Dose	How They Work	Side Effects
Common names include amlodipine, diltiazem and verapamil	1 time per day	Lower blood pressure by reducing the amount of calcium that enters the heart and blood vessels, so blood vessels widen and blood can flow easily	Headache and fluid retention are the most common side effects.

Diuretic (Fluid pill)

Generic Name	Dose	How They Work	Side Effects
Common names include furosemide, chlorthalidone and hydrochlorothiazide	1-3 times per day	Lower blood pressure by removing excess salt and water from the body	Frequent urination, dehydration, and dizziness are the most common side effects. Kidney function and electrolytes should be checked regularly by your provider.

Aldosterone Blocking Agents

Generic Name	Dose	How They Work	Side Effects
Common names include spironolactone and eplerenone	1-2 times per day	Lower blood pressure by reducing excess salt and water from the body	Increased potassium level is the most common side effect. Kidney function and potassium levels should be checked regularly by your provider. Spironolactone may also cause breast enlargement or tenderness.

Beta Blockers

Generic Name	Dose	How They Work	Side Effects
Names ends in "lol" (examples are metoprolol or carvedilol)	1-2 times per day	Lower blood pressure by making your heart beat slower and with less force	Fatigue, shortness of breath and fluid retention are the most common side effects. These side effects usually go away with time.

Other Medications

- Alpha blockers (such as doxazosin, prazosin, or terazosin)
- Direct renin inhibitors (such as aliskiren)
- Central alpha agonists (such as clonidine or methyldopa)
- Direct vasodilators (such as hydralazine or minoxidil)

Medications to Avoid

- Non-steroidal anti-inflammatory medications, known as NSAIDs such as ibuprofen (Advil®, Motrin®) or naproxen (Aleve®). A better alternative may be acetaminophen (Tylenol®) or trying physical therapy for pain.
- Decongestants such as pseudoephedrine (Sudafed®), phenylephrine, or oxymetazoline (Afrin®). A better alternative may be saline nasal spray.
- Herbal supplements and vitamins as they do not have the same regulations as prescription and over-the-counter medications. Also, they are usually not tested in humans, and the effects on the human body are not entirely known. A once-a day over- the-counter multivitamin is a better alternative.

Chapter 3. Diet, Weight Loss and Exercise

Making changes to your diet is a proven way to help control high blood pressure. These changes can also help you to lose weight and lower your chance of heart disease and stroke.

Your provider can refer you to a dietician who can help you create a healthy meal plan.

What can you do?

- Eat a heart healthy diet like the DASH Eating Plan
- Limit salty and processed foods and eating out
- Participate in daily exercise and maintain a healthy weight
- See a dietitian to learn more about eating heart-healthy meals

The DASH Eating Plan

The low salt Dietary Approaches to Stop Hypertension (DASH) eating plan is proven to lower blood pressure. The DASH eating plan focuses on fruits and vegetables, fat-free and low-fat dairy products. It includes whole grains, beans, fish, poultry, seeds, nuts and vegetable oils. It limits sodium, sweets, sugary beverages and red meats. The DASH eating plan is low in saturated and trans fats that are not good for your heart. It is rich in potassium, magnesium, fiber and protein. More information about the DASH eating plan is available at: https://www.nhlbi.nih.gov/health-topics/dash-eating-plan.

Speak with your provider about the DASH eating plan if you have chronic kidney disease because the DASH eating plan may not be appropriate for you.

Type of Fat	Foods with this type of fat	Included in the DASH plan?
Unsaturated fats	Soybean, canola, olive, and sunflower oil Liquid or soft tub margarines	Yes
Omega 3 fatty acids	Fatty, cold-water fish (salmon, tuna, mackerel, sardines) Flaxseed oil and ground flaxseed	Yes
Saturated fats	Animal fats (fatty meats, whole milk, butter, cream) Palm, palm kernel and coconut oil	No
***Trans* fats**	Hydrogenated oil used in foods such as fried foods, crackers, chips, baked goods	No

Limiting salt (sodium) in your diet

Salt (sodium chloride) is about 40% sodium. Eat less sodium to lower blood pressure. Eating too much sodium is a big cause of high blood pressure. The more sodium you eat, the higher your blood pressure will be.

How does sodium affect your heart and kidneys?

Where sodium goes, water goes. When you eat salty foods, the sodium goes into your blood stream and water follows it. You may notice swelling in your hands, feet or face. This extra

water raises the volume of your blood and causes your blood pressure to rise. Then your heart has to work harder to pump the extra amount of blood through your blood vessels. If your blood pressure is high, it can cause damage to your kidneys and heart.

How to Read a Food Label for Sodium Content

Packaged foods have a label that tells you the amount of sodium per serving in that food.

Look for the serving size at the top of the label.

Then, look for the amount of sodium in that serving of the food in the middle of the label.

If the food contains more than 200 mg of sodium, you should decrease the size of the serving or find a substitute for that food.

Nutrition Facts

Serving Size 1 cup (228g)
Servings Per Container 2

Amount Per Serving	
Calories 90	**Calories from Fat** 30
	% Daily Value*
Total Fat 3g	**5%**
Saturated Fat 0g	**0%**
Cholesterol 0mg	**0%**
Sodium 300mg	**13%**
Total Carbohydrate 13g	**4%**
Dietary Fiber 3g	**12%**
Sugars 3g	
Protein 3g	

Vitamin A 80%	•	**Vitamin C 60%**
Calcium 4%	•	**Iron 4%**

* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:

	Calories:	**2,000**	**2,500**
Total Fat	**Less than**	**65g**	**80g**
Sat Fat	**Less than**	**20g**	**25g**
Cholesterol	**Less than**	**300mg**	**300mg**
Sodium	**Less than**	**2,400mg**	**2,400mg**
Total Carbohydrate		**300g**	**375g**
Dietary Fiber		**25g**	**30g**

Calories per gram:
Fat 9 • Carbohydrate 4 • Protein 4

Low Sodium Guidelines

Sodium is found in many foods, especially in canned and processed foods. Decreasing the amount of sodium you eat will help keep extra fluid out of your body and your blood pressure from being too high. Not eating a large amount of sodium at one time will also help.

Experts disagree on the right amount of sodium in the diet, but a reasonable goal for most people with hypertension is to limit sodium intake to less than 2400 milligrams (mg) per day. An example of how to divide up sodium intake in one day is below:

- 700 mg per meal (3 meals per day)
- 200 mg per snack (1 snack per day)

Planning ahead, preparing most meals at home from fresh ingredients, reading food labels and eating balanced meals will help keep your sodium intake lower.

Most sodium comes from salt. Keep in mind that most of this salt is already added to prepared foods. Here are some ways to reduce the sodium in your diet:

AVOID TABLE SALT

1 teaspoon of table salt = 2300 mg sodium

This is almost the total amount you should have for the entire day!

AVOID PROCESSED FOODS

Examples include canned foods, frozen meals, snack foods, “instant” foods, packaged meals, most cheeses, deli meats, canned meats, pickles and pickled foods, cured or smoked meats, and pre-packaged ready-to-eat and boxed dishes. Some processed foods say “low sodium” or “low salt,” but may still have too much sodium. **CHECK THE LABEL!**

AVOID SAUCES AND SEASONINGS

Examples include bouillon cubes, broths, barbecue sauce, soy sauce, ketchup, and salad dressings. Some sauces and seasonings say “low sodium” or “low salt” but they may still have too much sodium. Be sure to check the label!

You can use salt substitutes. Some patients may need to limit how much they use salt substitutes because they contain potassium. If you have had problems with high potassium, you should not use salt substitutes. Ask your health care provider if you are not sure. Some examples of salt substitutes are McCormick® salt-free seasoning, Nu Salt®, and Morton’s Salt Substitute®. You can also use these herbs and spices in place of salt:

Pepper
Mrs. Dash®
Garlic/garlic powder
Oregano
Dill
Thyme
Lemon Pepper (no salt added)
Onion/onion powder
Rosemary
Cinnamon
Basil
Sage
Paprika
Turmeric
Dry mustard
Curry powder

Helpful Tips

- Choose fresh or frozen vegetables (without sauces) and fruits
- Use “No Salt Added” canned vegetables
- **Remember, many “low sodium” and “low salt” foods still have TOO MUCH SODIUM! (Read the label).**
- When eating out, ask for your meal to be cooked without salt
- Request sauces and salad dressings to be “on the side” so that you can use less
- Limit condiments such as ketchup, mustard, mayo)
- Limit eating out to once per week
- Avoid packaged, processed, and pre-made foods
- Get rid of the saltshaker and seasonings that have sodium in them

More information and tips for reducing sodium intake can be found in the “Additional Information” section of this booklet.

Losing Weight

Maintaining a healthy weight is important for blood pressure control. Even a small weight loss of 10 pounds can help to decrease blood pressure in people who are overweight. Ask your health care provider what a healthy weight is for you.

Weight Loss Tips:

- Eat 4-5 small meals per day.
- Listen to your body. When you are full, stop eating.
- Drink plenty of calorie-free drinks (water, unsweetened tea or coffee). You may be thirsty, not hungry.
- Choose lean meats, low-fat dairy (milk, low-sodium cheese, and yogurt).
- Eat high fiber foods including vegetables, fruits, beans and whole grains.
- Cut back on sugar including juice, soda, and sweets such as cakes, cookies and candies.
- Limit the amount of alcohol (beer, wine, and liquor) that you drink.
- Eat in one place only, such as at the kitchen table. Don't eat in the car, bedroom or while watching TV.
- Cook without adding fat (bake, broil, roast, boil). Use nonstick cooking sprays instead of butter or oil.
- Make half of your plate fruits and/or vegetables.
- Drink water while you cook and during meals.
- Use smaller plates, bowls, glasses, and serving spoons.
- Use measuring cups and spoons for proper portion control (see BELOW).
- Do not put serving dishes on the table. This will help you avoid a second portion.
- Make mealtime special by using pretty dishes, napkins, and glasses.
- Eat slowly. Put your fork down between bites. Cut your food one bite at a time.
- Enjoy fruit for dessert instead of cake, pie, or other sweets.
- Leave a few bites of each food on your plate.
- Remove your plate as soon as you have finished eating.

Talk with your health care provider about a healthy meal plan that is right for you!

In the "Additional Information" section of this booklet is a 3-day food log. Complete this prior to your next visit and review with the dietician or health care provider.

Portion Control for Weight Loss

Portion size matters. Research has shown that people consistently eat more food when offered larger sized portions. Therefore, portion control is important when you are trying to lose weight and keep it off. When measuring portion size, you do not need to carry around measuring cups. Instead, use your hand as a guide!

For example, 1 portion of meat or fish is 3 ounces. Three ounces is about the size of the palm of your hand. Likewise, 1 portion of cheese is about the size of your thumb.

Your hand makes it easy for you to imagine healthy portion sizes!

Exercise

Adults need at least 150 minutes of physical activity every week (this means on average about 22 minutes every day). Physical activity increases strength and endurance, makes your heart, lungs and bones stronger, helps you think and concentrate better, and helps with mood and self-esteem. It also lowers the risk of diabetes, high blood pressure and weight gain.

Talk with your provider before starting a new exercise program.

How to be More Active

- Limit TV, video games and computer time
- Be active with a friend or family member
- Change your activities to prevent boredom and burnout
- Schedule regular times for activities
- Have someone help keep you to your routine
- Use music during your activities
- Keep your walking shoes in your car or nearby at work

Suggested Activities

- Walking
- Biking
- Team sports
- Swimming
- Dancing
- Hiking
- Tennis
- Housework

Start out with small goals. You can begin with 20 minutes of exercise, 2 days per week, and then slowly increase the number of days per week and the length of time that you exercise. Be sure to warm up before you exercise and cool down after you exercise.

- Find a walking trail near you
- Check out activities at state parks
- Schedule time with friends and family members that involves physical activity
- Use a work-out video

Use the exercise journal in the “Additional Information” section of this booklet to track your activity.

“Cardio” = fast walking, swimming, biking, hiking, running, dancing. These exercises increase your heart rate, make you breathe deeply and use muscle groups.

Strength = weightlifting, resistance bands, and doing sit-ups. These exercises target muscles and help to build strength.

Start with 10 to 15 minutes per day and increase the time each week toward a goal of 30 to 60 minutes per day. Include both “cardio” and strength exercises in your fitness program.

The most important thing is to start moving more each day. Any increase in movement has positive health effects. The more the better.

Chapter 4. Measuring Blood Pressure

The only way to tell if you have high blood pressure is to have it checked. Checking your blood pressure is easy, quick and painless. Your blood pressure will be measured when you visit your health care provider, but you can also check your own blood pressure at home, or at a local pharmacy or grocery store.

…By your health care professional

To measure your blood pressure, your health care provider wraps a special cuff around your arm. Cuffs come in different sizes, and it will be important that the proper cuff size is used. The cuff is connected to a gauge that will measure your blood pressure. The cuff will be inflated to squeeze your arm, and then the air will slowly be released. The health care provider will listen to your pulse with a stethoscope and watch the blood pressure gauge. This gauge uses a scale called “millimeters of mercury” (mmHg) to measure the pressure in your blood vessels. Alternatively, an automated or digital device may be used.

Your provider will report your blood pressure as the systolic pressure over the diastolic pressure. Discuss with your provider what your target blood pressure level is. Your target (goal) blood pressure is:

_____/_____ mmHg

Your blood pressure can be affected by simple things like talking during the measurement or crossing your legs while the measurement is being taken. Before your blood pressure is measured in the office:

- Do not eat, smoke, or exercise for at least 30 minutes; do not use any medicine that can raise blood pressure (i.e. nasal sprays).
- Use the bathroom if you need to before getting your blood pressure checked.
- Rest for at least 5 minutes before your blood pressure is measured. Sit in a comfortable, relaxed position with both feet on the floor.
- Do not move or talk while the blood pressure is being measured.

Remember that your blood pressure changes throughout the day. Blood pressure is usually highest in the morning when you wake up and move around. Blood pressure usually decreases throughout the day, and is typically lowest at night. Blood pressure measurements may differ by as much as 10 to 20 mmHg between your right and left arm. Your health care provider will decide which arm to use for blood pressure measurements if that is the case (usually the arm with the higher readings).

Electronic devices are used more frequently to measure blood pressure in the office. There are specialized electronic devices that perform automated office blood pressure monitoring (AOBP). With AOBP, multiple blood pressure readings are recorded using a fully automated monitor while you are resting quietly and alone. Usually, three measurements are taken and the results are averaged. All the conditions noted above, in addition to proper cuff size and placement, are needed to ensure accurate readings with electronic devices.

…At your local pharmacy

Public blood pressure machines, such as those found in pharmacies, may provide helpful information about your blood pressure, but they may also have some limitations. The accuracy of these machines depends upon several factors, such as a correct cuff size and proper use of the machines. Ask your health care provider for advice on using public blood pressure machines.

…At home

How do you choose a home blood pressure monitoring device?

If you are buying your own blood pressure monitor to use at home, there are a few points to consider.

- For most patients, arm cuffs are most accurate, but wrist cuffs are a good choice if you can't find a cuff to fit your upper arm.
- Finger cuffs on blood pressure monitors are less accurate and should not be used.
- Monitors are available with larger displays that are easier to read.
- Always purchase a monitor that has the correct cuff size for your upper arm. Most cuffs have lines showing whether they are of the proper length when wrapped around the arm.

The recommended cuff size for accurate measurement of blood pressure is shown in the table below.

Arm Circumference	Recommended Cuff Size
22-26 cm	12 x 22 cm (small adult size)
27-34 cm	16 x 30 cm (regular adult size)
35-44 cm	16 x 36 cm (large adult size)

Ask your health care provider to measure your arm circumference.

Your arm circumference is:

_____ **cm**

The following devices are highly recommended by Consumer Reports and have been certified (confirmed to be accurate) by respected organizations. You can also check these websites to find devices that have been carefully tested: www.validatebp.org and www.stridebp.org. Bring your blood pressure monitor to your healthcare provider's office so it can be checked to be sure that it's accurate in you.

	Product	Price	Comments
Standard arm cuff	Omron 3 series Model # BP7100	Walmart: $32.84 Target: $32.99 Walgreens: $59.99 Amazon.com: $29.99	- Cuff fits standard and large arms - Stores up to 14 readings at a time - Requires 4 AA batteries
Fancy arm cuff	Omron 10 series wireless Model # BP7450	Walmart: $76.99 Walgreens: $99.99 Amazon.com: $69.86	- Automatically tests 3 consecutive readings one minute apart and calculates the average - Multi-colored BP level indicator lights - Stores up to 200 readings at a time - Wireless/Bluetooth connectivity - AC adapter included
Wrist cuff	Omron 3 series Model # BP6100	Walmart: $42.00 Target (7 series only): $58.99 Walgreens: $69.99 Amazon.com: $42.00	- Stores up to 60 readings at a time - Displays average of last 3 readings taken within 10 minutes - Fits wrists 5 ¼ inches to 8 ½ inches - Requires 2 AAA batteries (included)

Please note that the information in the table above, including pricing, is accurate as of October 2021.

How do I make sure I am measuring my blood pressure accurately?

Follow these steps to make sure that you are measuring your blood pressure accurately.

Before you take your blood pressure:

- Do not measure your blood pressure within 30 minutes of smoking, drinking alcohol, eating, or vigorous exercise. Do not take decongestants for 30 minutes before you take your blood pressure.
- Use the bathroom if you need to before taking your blood pressure
- Sit comfortably in a chair with your arm resting on a countertop or table at about the level of your heart.
- Sit with your back supported while keeping your feet flat on the floor.
- Remove clothing from the arm before applying the cuff.
- Rest for 5 minutes before taking your blood pressure.

When you are ready to take your blood pressure:

- Continue to sit with your back supported. Uncross your legs. Keep your feet flat on the floor.
- Follow the instructions for your device. Put the cuff on by wrapping it around your bare arm above your elbow. Face the palm of your hand up to relax your arm muscles.
- Rest your arm on a table or another flat surface at the level of your heart. Keep it stretched out and relaxed. Sit still.
- Do not talk while taking your blood pressure.
- Following the directions of the monitor you are using, press the button to start the machine. The cuff will inflate and slowly deflate by itself.

Recording your blood pressure:

- The machine will display two numbers. Write down both numbers, and the date and time of the measurement if the machine doesn't store that information automatically. If there is a pulse recorded on the display, write that down too. Record every measurement (even if you think it is incorrect!).
- Wait one minute and then repeat the steps. You should always check at least two measurements one minute apart and write them down. This is **one set** of blood pressure readings.
- Make **one set** of blood pressure readings in the morning before taking your medications and another in the late afternoon before supper. Do this one or more days each week and then every day during the week prior to your next visit to your health care provider. Set a time to measure your blood pressure- don't just do it when you feel poorly! And remember that blood pressure varies a lot from day to day and moment to moment, so don't be alarmed by occasional high or low readings.

Remember to bring your record of blood pressure readings and your blood pressure monitor to your clinic appointments. Your health care provider may periodically check the accuracy of your home blood pressure device.

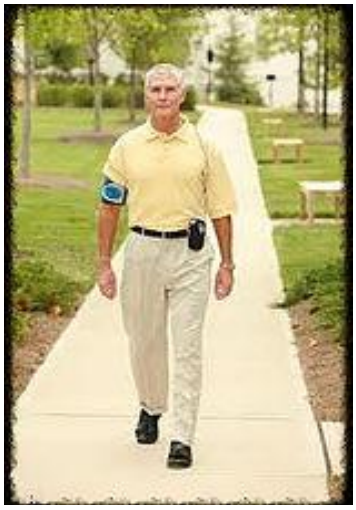
You may use this sheet to record your home blood pressure readings. You may not need to use it if your blood pressure device is able to store your readings, and you are able to share those readings with your health care provider.

In the “Additional Information” section of this booklet are additional blank sheets.

	MORNING			EVENING		
Date	Time	#1	#2	Time	#1	#2
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						
Day 6						
Day 7						

…Ambulatory blood pressure monitoring

Your health care provider may advise that you undergo Ambulatory Blood Pressure Monitoring (ABPM). ABPM is used to get a more accurate measurement of your average blood pressure during your usual daily activities and at night, or because you have different readings in the office compared to blood pressure readings at home, or for other reasons. You will wear a portable blood pressure monitor on your arm that will measure your blood pressure frequently over 24 hours. A report of your blood pressures during the day and night will be produced from the information obtained and stored in the recorder. Your health care provider will use this information to help manage your blood pressure.



Chapter 5. Additional Information

Additional information about high blood pressure can be found at the following websites. Please remember that the medical information that you find on these websites does not take the place of the recommendations of your health care provider.

- The American Heart Association (www.heart.org)
- The U.S. Center for Disease Control (www.cdc.gov/bloodpressure/)

Recipes

- http://www.heart.org (find recipes under getting healthy, nutrition center)
- http://www.mayoclinic.org/healthy-lifestyle/recipes/low-sodium-recipes/rcs-20077197
- www.MyPlate.gov
- www.Dashdiet.org

If you have kidney disease, then additional information can be found at the following websites:

- NKF Serving the Carolinas—North Carolina
  Division: NKF Serving the Southeast
  4701 Hedgemore Drive, Suite 810, Charlotte NC 28209
  https://www.kidney.org/offices/nkf-serving-carolinas-north-carolina
- The Renal Gourmet: What to Cook When Your Kidneys Quit! (http://kidneycookbook.com)

Recipes

- http://www.davita.com/recipes/
- Culinary Kidney Cooks (CKS)
- CKC published Cooking for David: A Culinary Dialysis Cookbook and Food Pyramid for Healthy Eating with Kidney Disease

  Phone: (714) 842-4684

  Website: www.culinarykidneycooks.com

MEDICATION LIST (Update and bring to each visit)

Medication Name	Strength	Dose	Frequency
Lisinopril	10 mg	1 tablet	Every morning

(Sample entry)

3-DAY FOOD AND BEVERAGE LOG (Complete prior to your next visit)

Day 1
Breakfast
Lunch
Dinner
Snack(s)

Day 2
Breakfast
Lunch
Dinner
Snack(s)

Day 3
Breakfast
Lunch
Dinner
Snack(s)

EXERCISE JOURNAL (Complete and review with your healthcare provider)

	MORNING			EVENING		
Date	**Time**	**#1**	**#2**	**Time**	**#1**	**#2**
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						
Day 6						
Day 7						

Foods Lower in Sodium That Can Be Included in Your Diet

Fruits	Fresh, frozen or canned fruits
Vegetables	Fresh, frozen or canned vegetables with no added salt
Proteins	Fresh, uncooked meats (chicken, beef, pork, fish, turkey); Eggs; Unsalted nuts and nut butters; Dried beans or canned beans with no added salt; Canned tuna, drained
Grains	Bread and tortillas; Plain pasta, rice and quinoa; Unsalted popcorn
Dairy	Low-fat milk and yogurt; Unprocessed cheese (cheddar, mozzarella, Swiss)
Other	Unsalted crackers, chips, pretzels; salt-free herbs and spices; Unsalted butter

Foods Higher in Sodium to Avoid in Your Diet

Fruits	None
Vegetables	Canned or pickled vegetables; Frozen vegetables in sauce; Vegetable juices
Proteins	Deli meats; Hotdogs; Bacon; Sausage; Precooked meats; Breaded meats; Frozen meats that are already seasoned
Grains	Instant oatmeal; Rice or pasta with a seasoning packet; Quick bread mixes and biscuits
Dairy	Cottage cheese; Buttermilk; Processed cheese /spreads
Other	Pre-made pies and cakes; Condiments such as ketchup, mustard, BBQ sauce, soy sauce, salad dressing and hot sauce

Lowering Sodium in Common Foods

Original	Better	Best
Peanut Butter 2 Tbsp=150 mg	Low-Sodium Peanut Butter 2 Tbsp=65-80 mg	Unsalted Peanut Butter 2 Tbsp=5 mg
Ham, Packaged or Deli 2 ounces=560 mg	Low Sodium Ham 2 ounces=450 mg	Cooked Fresh Ham 2 ounces=36 mg
Turkey, Packaged or Deli 2 ounces=540 mg	Low Sodium Turkey 2 ounces=260-420 mg	No Salt Added Turkey 2 ounces=25-65 mg
Flavored Instant Oatmeal 1 pkg=170-200 mg	Plain Instant Oatmeal 1 pkg=75 mg	Old Fashioned or Quick Oats ½ cup dry= 0 mg
French Fries Med=570 mg	French Fries No Salt Med=50-100 mg	½ Baked Potato Unsalted=0 mg
Mashed Potatoes Tub ½ cup=410 mg	Mashed Potatoes Dry (made with 2% milk) ½ cup=380 mg	Homemade Mashed Potatoes (no salt) ½ cup=75 mg
Canned Green Beans ½ cup=380 mg	Low Sodium Canned Green Beans ½ cup=10 mg	Fresh, Frozen or No Salt Added Canned Green Beans ½ cup=< 10 mg
Salad Dressing 2 Tbsp=280-330 mg	Low Sodium Salad Dressing Not available	Homemade Salad Dressing 2 Tbsp=<20 mg
American Cheese 1 ounce=230-460 mg	Cheddar Cheese 1 ounce=125-140 mg	Swiss Cheese 1 ounce=35 mg

Homemade Low Sodium Salad Dressings

Balsamic Vinaigrette (sodium < 10 mg per 2 Tbsp)

½ cup olive oil
¼ cup balsamic vinegar
1 tsp. Dijon mustard
1 tsp. honey
1 garlic clove

Ranch Dressing (sodium < 20 mg per 2 Tbsp)

2-6 ounce container Plain Greek Yogurt
2 tsp. dried dill
¼ tsp. orange peel or orange zest
¼ tsp. garlic powder
¼ tsp. onion powder
¼ tsp. onion flakes
¼ tsp. dry mustard

Making Good Choices When Eating Out

Restaurant	Less Healthy	More Healthy
McDonald's	Big Breakfast (regular biscuit) with small McCafe Mocha	Egg White Delight with small McCafe coffee
	1780 calories 59 g fat 1630 mg sodium	750 calories 7 g fat 740 mg sodium
Wendy's	Baconator®, medium natural-cut fries, small classic chocolate Frosty®	Grilled chicken sandwich, garden side salad with ranch dressing (no croutons), medium diet soda
	1720 calories 90 g fat 2450 mg sodium	470 calories 19 g fat 1170 mg sodium
Bojangles	2-piece dinner (leg & thigh), sweet potato pie, 22 oz. sweet tea	Grilled chicken sandwich, garden salad with buttermilk ranch dressing, 22 oz. unsweet tea
	1610 calories 92 g fat 3465 mg sodium	825 calories 52 g fat 1330 mg sodium
Hardee's	½ lb. Steakhouse Thickburger, medium crispy curls, chocolate hand-scooped ice cream malt®	Single all-natural burger, side salad, medium Coca-Cola Zero™
	2470 calories 141 g fat 4190 mg sodium	960 calories 63 g fat 1290 mg sodium
Subway	Footlong meatball marinara on 9-grain wheat with provolone cheese, cucumbers, green peppers, lettuce, tomato, red onion, regular mayonnaise, Doritos, 21-ounce sweet tea	6" oven roasted chicken on 9-grain wheat with Swiss cheese, cucumbers, green peppers, lettuce, tomato, red onions, yellow or deli brown mustard, baked Lays (plain), 21-ounce unsweet tea
	1690 calories 81 g fat 2560 mg sodium	510 calories 12 g fat 960 mg sodium

Nutritional facts were obtained from the website of each company on December 15, 2015. This table is for information only and is not a recommendation or endorsement for any of these restaurants.

LOG SHEET FOR RECORDING HOME BLOOD PRESSURE MEASUREMENTS

	MORNING			EVENING		
Date	Time	#1	#2	Time	#1	#2
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						
Day 6						
Day 7						

COMMENTS, CONCERNS, NOTES